

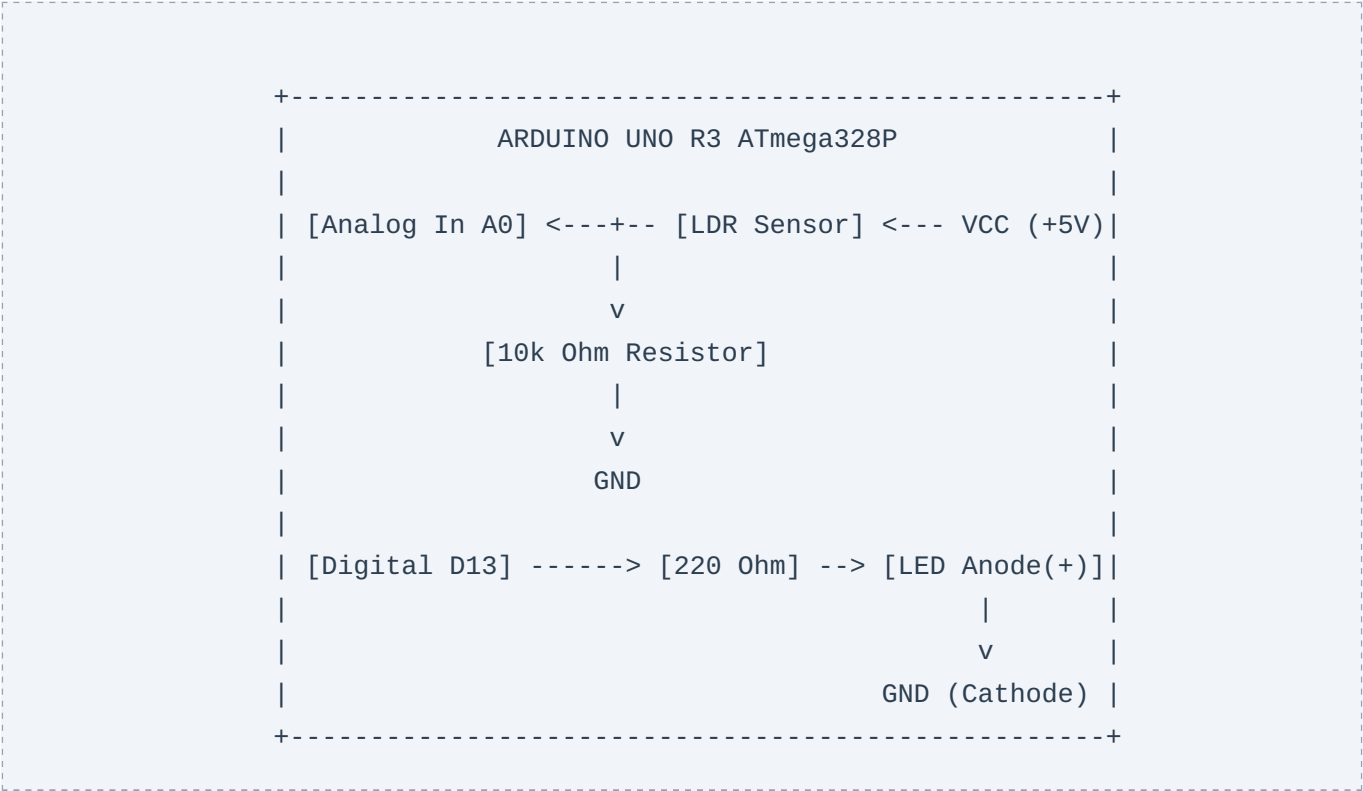
TASK 2: Sensor-Based Simulation

Automated Environmental Surveillance Loop & Firmware Design

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Objective: To design, simulate, and analyze an intelligent micro-controller automated environmental surveillance loop using Tinkercad/Proteus paradigms. The simulation integrates an analog ambient photo-transducer (Light Dependent Resistor) and a digital signaling indicator sub-system to mimic intelligent energy-saving and safety-critical threshold responses.

1. System Schematic Blueprint (Conceptual Representation)



2. Simulation Source Code (Embedded C / Arduino Framework)

```
// CodeAlpha IoT Internship
// Task 2: Sensor-Based Simulation
// Target Hardware: ATmega328P / Arduino Uno

const int LDR_PIN = A0;      // Analog pin connection for Light Dependent Resistor
const int LED_PIN = 13;      // Built-in indicator LED driven by digital line 13
const int THRESHOLD = 400;   // Calibrated luminescence baseline transition threshold

void setup() {
    pinMode(LED_PIN, OUTPUT); // Configure digital channel 13 as a low-impedance output
    pinMode(LDR_PIN, INPUT);  // Establish A0 line as a high-impedance analog input receiver
    Serial.begin(9600);       // Initialize UART serial bus at a standard 9600 baud rate
}

void loop() {
    // Read raw 10-bit integer scalar (Range: 0 to 1023) representing sensor voltage division
    int rawLightValue = analogRead(LDR_PIN);
    Serial.print("Ambient Luminescence Raw ADC Output: ");
    Serial.println(rawLightValue);

    // Conditional comparison matrix determining state execution
    if (rawLightValue < THRESHOLD) {
        digitalWrite(LED_PIN, HIGH); // Assert high signal level to energize the indicator LED
        Serial.println("==> CRITICAL STATUS: Low Light / Obstruction Detected! LED ACTIVATED.");
    } else {
        digitalWrite(LED_PIN, LOW); // Pull line low to terminate current loop through LED
        Serial.println("==> STATUS NORMAL: Adequate Ambient Exposure. LED DEACTIVATED.");
    }
    delay(500); // Microcontroller pacing latency configuration to mitigate telemetry overflow
}
```

3. In-Depth Simulation Behavior & Execution Explanation

- **ADC Transduction Process:** The LDR sensor acts within a fixed pull-down resistor configuration. As ambient illumination drops, LDR resistance increases exponentially, causing the voltage across the analog pin A0 to drop toward ground level. The microcontroller's internal 10-bit register maps this voltage directly to an integer value between '0' and '1023'.
- **Hysteresis & Control Logic:** When the illumination drops below the internal setpoint ('400'), it triggers an immediate logic transition. The `digitalWrite(LED_PIN, HIGH)` command applies a 5V potential across pin 13, illuminating the diagnostic LED. In a real-world scenario, this logic translates directly to automated street lighting systems or security perimeter infraction alarms.